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ORIGINAL RESEARCH

Research on stock trend prediction method based on optimized random forest

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Abstract

As a complex hot problem in the financial field, stock trend forecasting uses a large amount of data and many related indicators; hence it is difficult to obtain sustainable and effective results only by relying on empirical analysis. Researchers in the field of machine learning have proved that random forest can form better judgements on this kind of problem, and it has an auxiliary role in the prediction of stock trend. This study uses historical trading data of four listed companies in the USA stock market, and the purpose of this study is to improve the performance of random forest model in medium- and long-term stock trend prediction. This study applies the exponential smoothing method to process the initial data, calculates the relevant technical indicators as the characteristics to be selected, and proposes the D-RF-RS method to optimize random forest. As the random forest is an ensemble learning model and is closely related to decision tree, D-RF-RS method uses a decision tree to screen the importance of features, and obtains the effective strong feature set of the model as input. Then, the parameter combination of the model is optimized through random parameter search. The experimental results show that the average accuracy of random forest is increased by 0.17 after the above process optimization, which is 0.18 higher than the average accuracy of light gradient boosting machine model. Combined with the performance of the ROC curve and Precision–Recall curve, the stability of the model is also guaranteed, which further demonstrates the advantages of random forest in medium- and long-term trend prediction of the stock market.

KEYWORDS

ensemble learning, finance, random forest, random search, technical indicator

1 | INTRODUCTION

Based on the long-term investment behaviour, it is more likely to achieve higher return rate by using a reasonable way to hold stocks than using other conservative investment behaviours. However, to grasp the trend of stock prices requires a large number of factors, which will become an extremely difficult task when it comes to analysing and forecasting stocks.

Many commercial companies and research institutions have developed stock predictors based on statistical models, and it is expected to obtain more regular information in the analysis of historical data or directly assist decision making; but the stock market is dynamic, non-linear and complex, therefore it is still very difficult to predict stock trend effectively. Mandelbrot proposed that normal distribution cannot explain the yield sequence of stocks reasonably, and many abnormal properties in financial transactions cannot be explained based on the effective market hypothesis [1]. But the invalid market proposed by Benjamin Graham gives the possibility and practical significance to stock forecasting. The two directions of quantitative finance are intelligent timing and intelligent stock selection. The essence of these two directions is to predict the rise and fall of stocks. At the same time, quantitative investment and machine learning are more and more integrated [2–6], and the research of stock

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forecasting using technical indicators and machine learning algorithm is developing rapidly.

In order to predict the price trend of the stock, this study uses the random forest model, the data source is Yahoo financial interface. Among them, obtained stock data is reprocessed by exponential smoothing method, and the processed data are used to calculate the technical indicators. In this study, D-RF-RS method is proposed in the stage of feature screening and model training. Preliminary technical indicators are sorted by the importance of decision tree features, and the feature set obtained in this way is used as the input to the model. At the same time, the random search is utilized to optimize the parameters of the random forest model. The goal of this study is to evaluate the performance of random forest in medium- and long-term stock forecasting, improve the accuracy of stock forecasting, and assist relevant financial practitioners to judge the tendency of stock rise and fall.

Many researchers have used support vector machine, random forest and other machine learning models to predict the stock trend. In many application areas, integration models have played an important role [7–10]. But it is still challenging to deal with time series data, select technical indicators and optimize the combination of parameters. Few research studies systematically build stock forecasting model according to this complete process, hence it is of great significance to apply this method used in this study to stock market forecasting.

The rest of this paper is arranged as follows: The second section is the summary and analysis of relevant research work. The third section is the introduction of the model, which explains in detail the principle of using the model and its applicability in this study; The fourth section is the experimental process, including data selection and processing, feature extraction, parameter optimization, introduction of the evaluation criteria and other experimental stages; The fifth section is the explanation and analysis of the experimental results. The sixth section is the conclusion.

2 | RELATED WORKS

Many researchers use different technologies to classify and predict the future value of stocks, including statistics, data mining and so on.

Tan et al. [11] takes stock selection as the research direction, uses the stock information of Chinese stock market as the data source, uses the fundamental/technical feature space and pure momentum space combined with random forest to predict short-term and long-term share price trend; standardization of fund performance evaluation index were 2.75 and 5, showing the effectiveness of the model to choose a strategy.

Kofi et al. [12] studied macroeconomic variable factors and found that the application of more important features to train the random forest model reduced the prediction error by 7.1% compared to the application of all feature training models, indicating the positive effect of screening macroeconomic factors on stock market forecasts.

In the above two studies, the selection of the feature set is an important step.

Ballings et al. [13] studied the traditional model and the integrated model in the field of machine learning, and proved that the integrated model is better than the single model in predicting the financial data based on time series. The random forest with bagging (bootstrap aggregating) is one of the excellent integrated models, which can prevent the over fitting phenomenon in the training process. Basak et al. [14] trained random forest and XGBoost by using exponential smoothing data, and the trend prediction accuracy of these two classifiers increased with the improvement of the time window. The overall experiment shows that random forest has more advantages than XGBoost. Based on the technical indicators of the stock market, Luckyson et al. used the random forest model to predict the stock trend, and the experiment shows that the random forest is better than the support vector machine and the logical regression, which can obtain more effective trend prediction results. Yanjun Chen constructed a financial transaction strategy model based on LightGBM to solve the problem of sparse high-dimensional feature matrix in financial data. Compared with OpenGL Mathematics, deep neural networks and support vector machines models, the prediction error of LightGBM model is significantly reduced and the prediction precision is higher [15]. SVM incorporates structured risk minimization to decrease errors and to improve the effectiveness of classification.

In the study of Manik et al, the intraday stock status is mined using ten different classifiers, including C4.5, random forest, logistic regression, linear discriminant, SVM, quadratic SVM, cubic SVM, Gaussian SVM and so on. The performance of different classifiers is evaluated on the basis of various indexes such as accuracy, misclassification rate, precision, recall and so on [16]. Decision tree performs better when there are discrete features [17].

3 | MODEL INTRODUCTION

Random forest is a classical ensemble learning model, and its weak learner is decision tree model. Based on the basic model, random forest can be classified as bagging-integrated learning model, which is an extended variant of Bagging. Decision tree is a widely used white box classification algorithm, which has the characteristics of visualization, while random forest is a black box algorithm, which is unexplainable.

Compared with solving the complex pre-problems, it is more meaningful to refine the rules into a series of simple problems. In the greedy algorithm, it is the best solution to place the problem with the largest information in front of this sequence. Based on this, the problem is added with a hierarchical structure. The gradual division of the amount of information naturally involves the decision tree.

The decision tree has two main parts, one is the quantitative measurement of purity, the other is the type of each node's problem.

The construction process of the tree should follow the “simple” principle, that is, the expected decision tree is simple

and compact. Occam razor principle also proposes that the simplest model that can interpret data is the best model. The purity of a subset can be measured by information gain or Gini impurity. Draw samples from a set corresponding to an internal node or leaf node, where the probability $\Pr(y)$ of y typical instance is proportional to the proportion of the type instance in the set. The Shannon entropy of labelled probability distribution is used to measure the statistical uncertainty of the obtained class:

$$H(\mathrm{Y}) = -\sum_{y\epsilon \mathrm{Y}} \Pr(\mathrm{y})\log \Pr(\mathrm{y}) \tag{1}$$

In the information gain method, the entropy of class probability distribution is used to measure the chaotic degree of a set. The following formula uses $\log_2 \Pr(y)$ to represent the unit of information (entropy) as binary bits.

$$\mathrm{H}(\mathrm{Y}) = -\sum_{y\epsilon \mathrm{Y}} \Pr(\mathrm{y})\log_2 \Pr(\mathrm{y}) \tag{2}$$

When a set is evenly divided by all kinds of instances, the entropy reaches the maximum value; when all the instances belong to the same class, the entropy takes the minimum value.

Information gain is represented by the change of entropy at the beginning and end of data set partition. The instance set is represented by D, and $D = D_{\mathrm{T}} U D_T$ is a partition, and the information gain can be expressed as follows:

$$IG = H(D) - \frac{|\mathrm{D_T}|}{\mathrm{D}} H(\mathrm{D_T}) - \frac{|\mathrm{D_F}|}{\mathrm{D}} H(\mathrm{D_F}) \tag{3}$$

Gini impurity (GI) is the frequency at which randomly selected elements from a set are incorrectly labelled, as follows:

$$GI(\mathrm{f}) = \sum_{i=1}^{\mathrm{m}} f_{\mathrm{i}}(1 - f_{\mathrm{i}}) = \sum_{i=1}^{\mathrm{m}} f_{\mathrm{i}} - \sum_{i=1}^{\mathrm{m}} f_i^2 = 1 - \sum_{i=1}^{\mathrm{m}} f_{\mathrm{i}}^2 \tag{4}$$

where m is the number of categories and f_i is the proportion of elements marked i in the collection. Estimating probability by frequency, that is, $\mathrm{f}_i \approx \mathrm{p_i}$ in data set D, the following results can be obtained:

$$GI(D) = 1 - \sum_{\mathrm{i}=1}^{\mathrm{m}} \mathrm{p}_{\mathrm{i}}^2 \tag{5}$$

In the field of machine learning, for classification problems, the output is usually determined by the majority vote, while for regression problems, the output is determined by the average value. The total number of input variables is n, and the construction process of each tree is as follows: selecting n' input variables to determine the division of a node, and n' is generally much smaller than n. By allowing repeated sampling, a tree is generated one at a time. At the same time, those instances that are not in the extraction range can be used to evaluate the error of the tree. For each node in the tree, n'variables are randomly selected, which are the basis for making decisions on this node. According to the purity measurement method (IG, GI), one type variable is selected to divide a node at a time. Subsets of these types can be randomly selected and a replacement variable is defined. When the type value is in this subset, it is 1, otherwise 0 is taken.

LightGBM model is a kind of tree based on gradient lifting learning framework [18, 19]. It was proposed by Microsoft Asia Research Institute in 2017, which is fully called Light Gradient Boosting Machine. It improves the disadvantages of traditional Gradient Boosting Decision Tree in terms of slow training speed and long-term. Light gradient boosting machine (LGBM) model is one of the most advanced gradient lifting algorithms. The model uses integrated gradient enhancement to improve the robustness of classifier, uses decision tree with limited depth to avoid over fitting, uses Gradient-based One-side Sampling to represent the importance of data instances, and uses histogram algorithm to find the best split.

4 | EXPERIMENTAL PROCESS

4.1 | Data selection and preprocessing

In order to form an effective prediction of the future medium and long-term market, this study selects the data to a certain time span as the training and testing data set of the model. At the same time, it is hoped that the larger data set can be used to select the input characteristics suitable for the medium and long term to effectively predict the future stock dynamics. In the experiment, historical stock data of Integrated Electronics Corporation (stock code: INTC), Google (stock code: GOOG), Advanced Micro Devices Inc. (stock code: AMD) and Activision Blizzard Inc. (stock code: ATVI) from 3 January 2015 to 20 October 2020 is used as research data, and a variety of technical indicators are obtained by using these data. Through the pandas-datareader package, the stock price data of Yahoo Finance is obtained as the preliminary data source of this study. The historical data of stocks is based on the time series [20–23] and the step size is 1 day. The data dimensions collected include opening price, closing price, highest price, lowest price and trading volume.

Exponential smoothing is often used in forecasting models. On the one hand, it takes the moving average method as the cornerstone, and discards the disadvantage of not adopting long-term data. On the other hand, it is compatible with some characteristics of the whole period average for data processing.

In this process, data with short time distance are given greater weight, while the data with long time distance are given index decreasing weight. The exponential smoothing of series Y can form a recursive representation:

$$\mathrm{ES}_0 = \mathrm{Y}_0, \quad \mathrm{t} = 0 \tag{6}$$

$$ES_t = \alpha * Y_t + (1-\alpha) * ES_{t-1}, t > 0 \quad (7)$$

Among them, ES_t is the smoothing value at time t, Y_t is the actual value of time t, and α (the value range is (0, 1]) is the smoothing constant. The larger value of α, the lower smoothness degree. When $\alpha = 1$, the smoothed data is equal to the actual data, that is, $ES_t = Y_t$.The value of α is a key step. From experience, if the data fluctuates too much, the value should be larger. Therefore, the value of α in this study is 0.7. This smoothing method can reduce or eliminate the random changes or noises in the historical data, so that the model can better identify the dynamic changes of the stock price in the medium and long term. The technical indicators are calculated from the time series data of exponential smoothing (exponential smoothing can be regarded as a special weighted moving average method in essence) and included in the feature matrix.

Based on Close (the closing price), the price rise and fall are calculated as follows:

$$pred_n = Label(Close_{n+h} - Close_n) \quad (8)$$

When $pred_n$ value is +1, it indicates that the price on h day presents a positive trend; when $pred_n$ value is −1, it indicates that the price on h day presents a negative trend. Finally, the stock data as numerical data through the rise and fall categories into nominal data.

4.2 | Feature extraction

The technical index used in technical analysis is an important parameter calculated from the stock data to predict the trend of the stock market. It is also a tool often used by investors. Technical indicators [24–28] are usually used as indicators of short-term stock price dynamic evaluation, and can also have a good effect in medium- and long-term use, such as the search for entry point and exit point. In order to verify the effectiveness of the process optimization stochastic forest model proposed in this study, the important technical representative indicators are taken as input characteristics, and the model is trained. Finally, up or down is taken as the output variable to predict the dynamic trend of a single stock. First of all, based on the purpose of medium and long-term stock forecasting, various indicators for technical analysis are roughly selected, as shown in Table 1.

In order to select appropriate technical indicators to establish an effective technical data system, the INTC stock data is selected as the data set, and feature importance analysis [15, 29] is performed through the construction of three decision trees to obtain the technical feature importance data as shown in Figure 1. importancei (i = 1, 2, 3) represents the feature importance obtained by the i-decision tree construction:

Retaining the technical features with a degree of importance exceeding 0.04, ten technical indicators of Donchian, MFI, OBV, ATR, TSI, Coppock, SO%d, ULTOSC, BOLL and Mass Index were finally selected as input features.

In this study, the overall data set of technical indicators obtained above is divided into two parts, two thirds of which are training set and one third is test set.

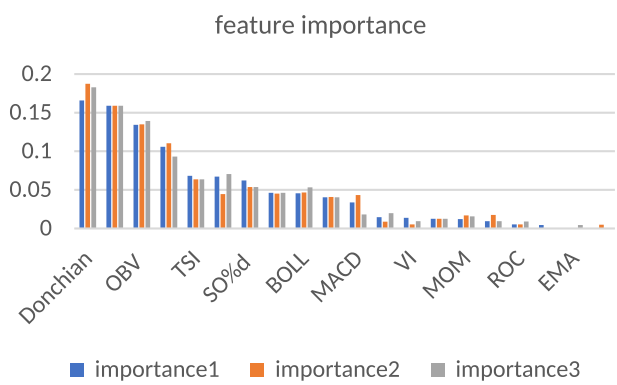

FIGURE 1 The importance ranking of features obtained from the construction of cubic decision tree

TABLE 1 Initial selected technical indicators

Technical indicators	Full name of index	Technical indicators	Full name of index
EMA	Exponential moving average	BOLL	Bollinger bands
MACD	Moving average convergence divergence	OBV	On balance volume
MTM	Momentum index	MA	Moving average
RSI	Relative strength index	TSI	True strength index
ATR	Average true range	Coppock	Coppock curve
ADOSC	A/D oscillator	SO%d	Stochastic oscillator
CCI	Commodity channel index	Mass index	Mass index
MFI	Money flows indicators	EMV	Ease of movement value
ULTOSC	Ultimate oscillator	VI	Vortex indicator
Donchian	Donchian channel	Chaikin	Chaikin oscillator

4.3 | Parameter optimization

Superparametric optimization [30] is a combinatorial optimization process. The main drawback of manual search is repetitive work and low efficiency. Taking INTC stock as an example, when the prediction window is 9 days and only the parameter n_estimator (number of trees) is adjusted manually, the changes of precision, recall rate, F1 value and accuracy rate are shown in Table 2 and Figure 2.

Grid search is a good super parameter search algorithm that is widely used and easy to implement in parameter optimization of random forest. It searches the best parameter value in the set based on the points within the range parameters. By using a wide range of parameters and using a smaller step size every time the parameters change, it is possible to obtain the optimal value under these conditions. When using a specific learning model, it can be observed that the importance of different subspaces of different datasets is different through several similar datasets (from different distributions). A grid with enough granularity to optimize the super parameters is also inefficient because the search dimension is exponential in the process of grid search, and the search dimension is irrelevant of the specific data set. In actual operation, grid search consumes much time and occupies a lot of resources. Therefore, this study adopts another method, namely random search, to obtain sample points in a random manner within a certain search range.

Random search strategy: (1) random sampling in the distribution; (2) equal probability sampling in the list; (3) traversing the sampling results obtained in the above two steps.

TABLE 2 The relationship between the number of trees and each evaluation index

n_estimator	Precision	Recall	F1 score	Accuracy
5	0.62	0.52	0.57	0.52
11	0.67	0.62	0.64	0.58
49	0.65	0.67	0.66	0.58
69	0.65	0.7	0.67	0.59
89	0.66	0.7	0.68	0.6
99	0.67	0.72	0.7	0.62
111	0.67	0.72	0.69	0.61
189	0.66	0.73	0.69	0.61

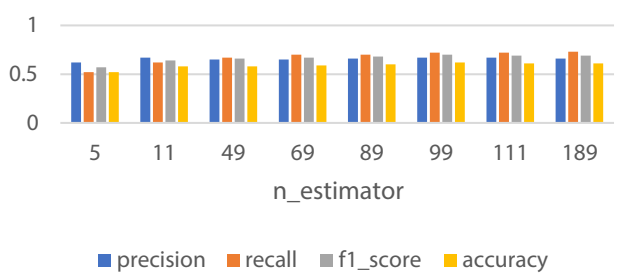

FIGURE 2 The relationship between the number of trees and each evaluation index

Random search is based on low effective dimension. Random search has the same efficiency in correlation subspace, which is equivalent to that it is only used to search related dimensions. Random search is more valuable than grid search. On the one hand, it can maintain reasonable efficiency, on the other hand, it has the advantages of simple and reproducible pure grid search.

In the actual experiment, the random number seed is also a parameter, but the fixed value should be used to avoid the influence of sampling error. The parameters such as the number of trees that the random forest needs to build determines the process of model optimization. The random forest model is essentially random. Setting different random states (specifically, random state is not set random_state parameter) can completely change the construction process. Such changes include each generated data set, each split training set and test set, and built model.

The number of trees in random forest is not the more the better. On the one hand, it is a matter of resources. On the other hand, when the number of trees increases to a certain extent, the performance of the model is basically stable. In addition, the random characteristics of the random forest make the model more robust to outliers, and the generalization ability is stronger. Too many trees will offset the advantages brought by the random characteristics.

The search range of random sampling in this study is summarized as a list of multiple parameters, and equal probability sampling is performed in the given list, as shown in Table 3.

This study uses random sampling to optimize the parameters of the random forest, uses 3-fold cross-validation, uses the AUC score of the ROC curve as the model evaluation index, and obtains the best number of decision trees (n_estimators), maximum tree depth (max_depth), and internally the minimum number of samples required for node subdivision (min_samples_split) and the minimum number of samples for leaf nodes (min_samples_leaf).

Using random sampling to optimize the parameters, it is found that the optimal parameters and the results before and after optimization are shown in Table 4.

4.4 | Evaluating indicator

For the purpose of trend prediction, this study adopts the simplest form of classification problem, that is, it is divided into two categories. Finally, the classification result of whether the stock trend has positive development is determined as + 1 or – 1. If the actual value is +1, and the model classification is also +1, the number of such data is recorded as tp; If the actual value is +1 and the model classification is −1, the number of such data is recorded as fn; If the actual value is −1 and the model classification is also −1, the number of such data is recorded as tn; If the actual value is −1 and the model classification is +1, the number of such data is denoted as fp. Based on the above four data, the confusion matrix is constructed as shown in Table 5.

TABLE 3 Range of parameter values in random search

Meaning	Parameter	Parameter value
The number of trees in the forest	n_estimators	9, 50, 99, 130, 180, 230
The maximum depth of the tree	max_depth	10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, None
The minimum number of samples required to split an internal node	min_samples_split	2, 5, 10, 20
The minimum number of samples required to be at a leaf node.	min_samples_leaf	1, 2, 4, 10, 30

TABLE 4 Comparison of experimental results before and after parameter optimization

	n_estimators	max_depth	min_samples_split	min_samples_leaf	Accuracy
Default	5	None	2	1	0.52
Optimal	130	90	20	10	0.61

Based on the above classification, different evaluation indexes of the model are derived. Accuracy refers to the probability of correct overall judgement of the classification model; Precision represents the probability that a sample predicted to be positive class is actually positive class, and recall, also known as sensitivity, represents the probability of the sample of actual positive class predicted as positive class. Specificity represents the probability of the sample of actual negative class predicted as negative class. 1-Specificity represents the probability of the sample of actual negative class predicted as positive class. F1 Score is the harmonic average of the model accuracy and recall rate. The calculation formula is as follows:

$$Accuracy = \frac{tp + tn}{tp + tn + fp + fn} \tag{9}$$

$$Precision = \frac{tp}{tp + fp} \tag{10}$$

$$Recall = Sensitivity = \frac{tp}{tp + fn} \tag{11}$$

$$Specificity = \frac{tn}{tn + fp} \tag{12}$$

$$1 - Specificity = \frac{fp}{tn + fp} \tag{13}$$

$$F1\ score = 2\frac{Precision \times Recall}{Precision + Recall} \tag{14}$$

5 | EXPERIMENTAL RESULTS

According to the optimal combination of parameters obtained through parameter optimization, the model is used to find the optimal value of prediction probability under different time windows. When the time window reaches 68 days, the precision is 0.88, the recall rate is 0.91, and the F1 value is 0.89, The accuracy rate was 0.89, and the prediction accuracy showed a dynamic change with time in the test set, and showed good prediction ability in 62–82 days, indicating that the optimized random forest model has more advantages for medium and long-term prediction, as shown in Figure 3.

TABLE 5 Confusion matrix

		Actual value	
Confusion matrix		**1**	**−1**
Predicted value	1	tp	Fp
	−1	fn	tn

In May 2020, Yanjun Che and others built a LightGBM model based on financial transaction strategy. The research results show that compared with the traditional linear model, LightGBM model has higher accuracy in dealing with financial data. In order to further verify the classification performance of random forest (D-RF-RS) after feature importance screening and random parameter optimization, this study constructs an optimized random forest stock prediction model and compares the empirical results of stock trend prediction with original random forest and LightGBM model, respectively. Figures 4–6 show the ROC curves drawn from the training results of the original random forest, the LGBM model, and the random forest after feature selection and parameter optimization.

According to the comprehensive comparison in Table 6, the performance of the optimized random forest model proposed in this study is better than that of the original random forest model and LGBM model. Therefore, the random forest algorithm with decision tree feature extraction and random parameter optimization has stronger generalization ability and better trend prediction ability.

In order to have a deeper understanding of the performance of the optimization method, this study uses the Precision–Recall curve to compare the performance of D-RF-RS model and LGBM model.

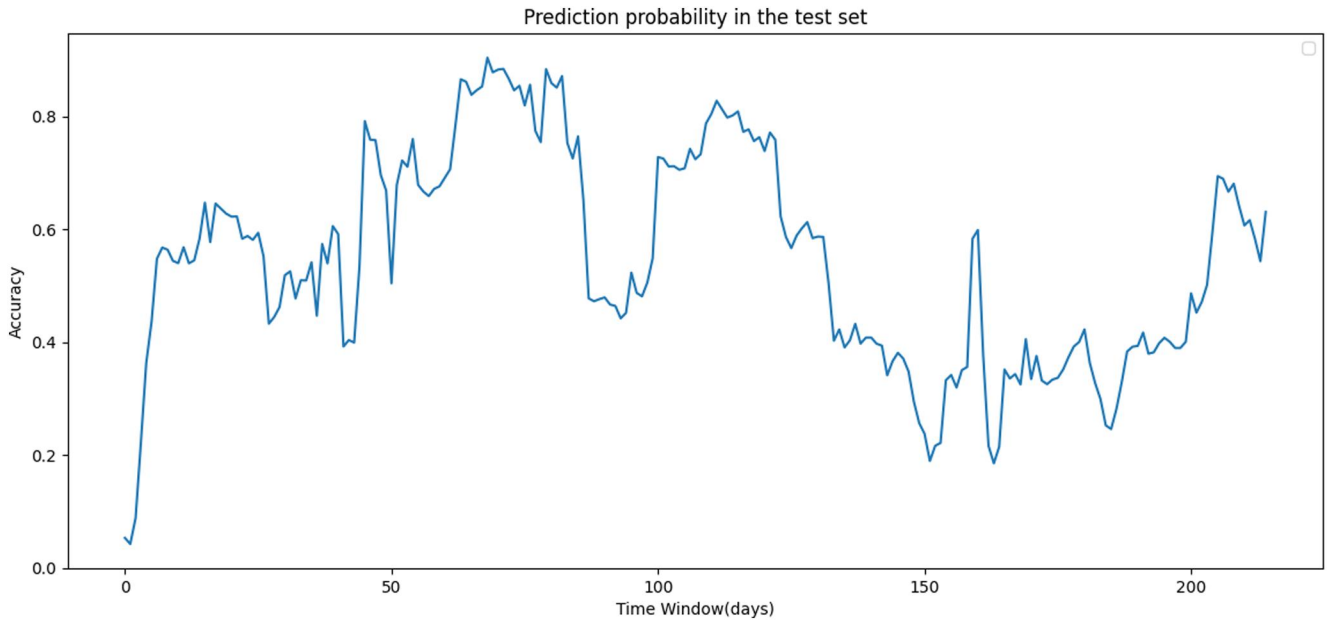

FIGURE 3 Relationship between prediction accuracy and time

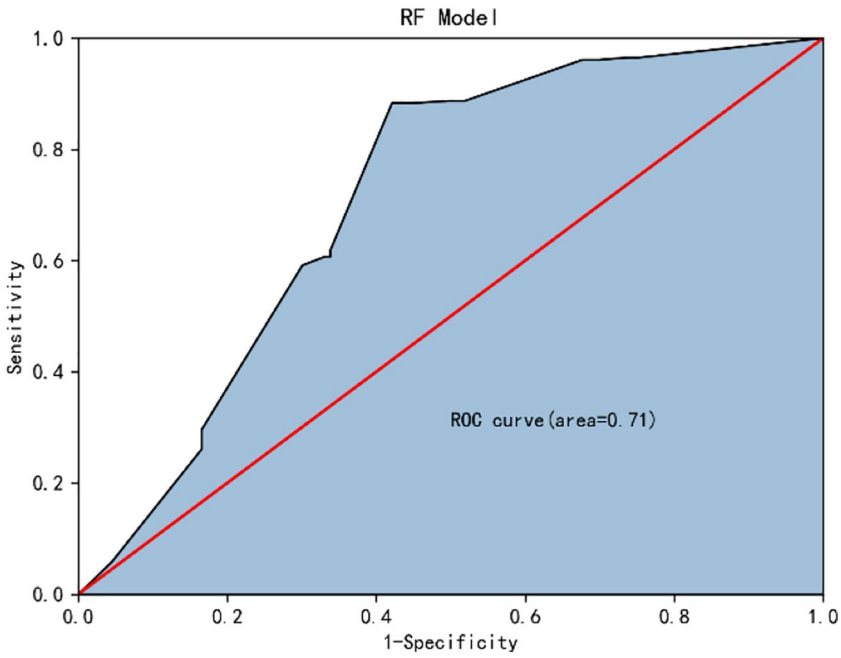

FIGURE 4 RF model ROC curve

The ROC curve mentioned in the text is used as a performance indicator, and its robustness can be better guaranteed when the positive and negative sample sizes are roughly the same. The Precision-Recall curve (PR curve) makes up for the deficiency of ROC curve in reflecting the model performance under the imbalance of positive and negative sample sizes. The combination of the two can save us from the need to pay attention to the changes of positive and negative sample sizes and play a key role in the comprehensive evaluation model. By comparing the performance of four stocks in the D-RF-RS model and LGBM model, that is, comparing Figures 7a and 8a, Figures 7b and 8b, Figures 7c and 8c, Figures 7d and 8d, respectively, it is concluded that the D-RF-RS model is more stable in stock forecasting.

6 | CONCLUSION

In order to further enhance the accuracy of stock trend prediction, this study uses the random forest model and optimizes the various processes of stock research to achieve the effective improvement of accuracy. This study selects the

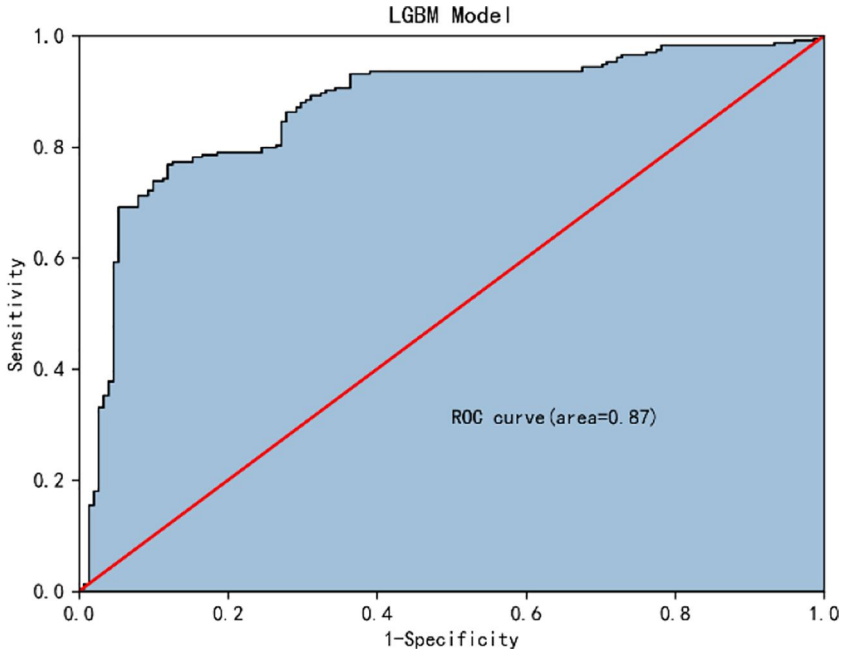

FIGURE 5 LGBM model ROC curve. LGBM, light gradient boosting machine

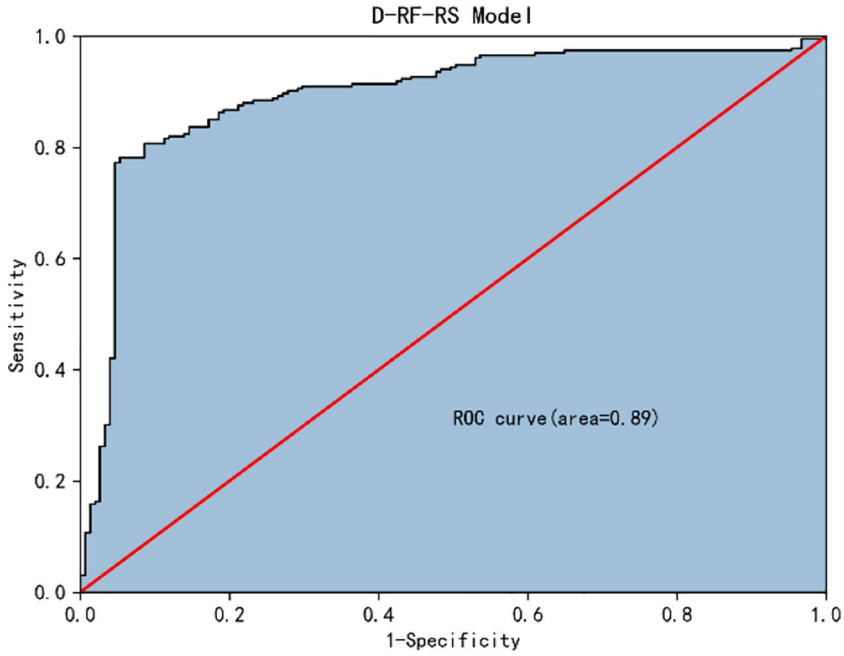

FIGURE 6 D-RF-RS ROC curve

stock data of four American listed companies in recent 5 years and uses exponential smoothing to preprocess the data. In the feature extraction, considering the medium- and long-term trend prediction factors, after the decision tree feature importance screening, ten technical indicators are adopted as the characteristics of stock prediction, which are applied to the random forest with parameter tuning through random sampling. From the analysis of specific data, compared with the original random forest, the accuracy of the optimized random forest on the four stocks is improved by 0.30, 0.07, 0.23 and 0.09, respectively. Compared with LGBM model, the accuracy of the optimized random forest on the four stocks is improved by 0.28, 0.02, 0.18 and 0.18, respectively. The results show that the above optimized random forest is more suitable for medium- and long-term prediction, and is better than the original random forest and the LightGBM model based on decision trees. Combined with the performance of ROC curve and Precision–Recall curve, it shows that the method used in this study is reliable.

TABLE 6 Experimental results of prediction accuracy of different models

Stock code	Primary random forest	LGBM model	Optimized random forest
INTC	0.59	0.61	0.89
AMD	0.69	0.74	0.76
ATVI	0.65	0.70	0.88
GOOG	0.68	0.59	0.77

Abbreviations: LGBM, light gradient boosting machine.

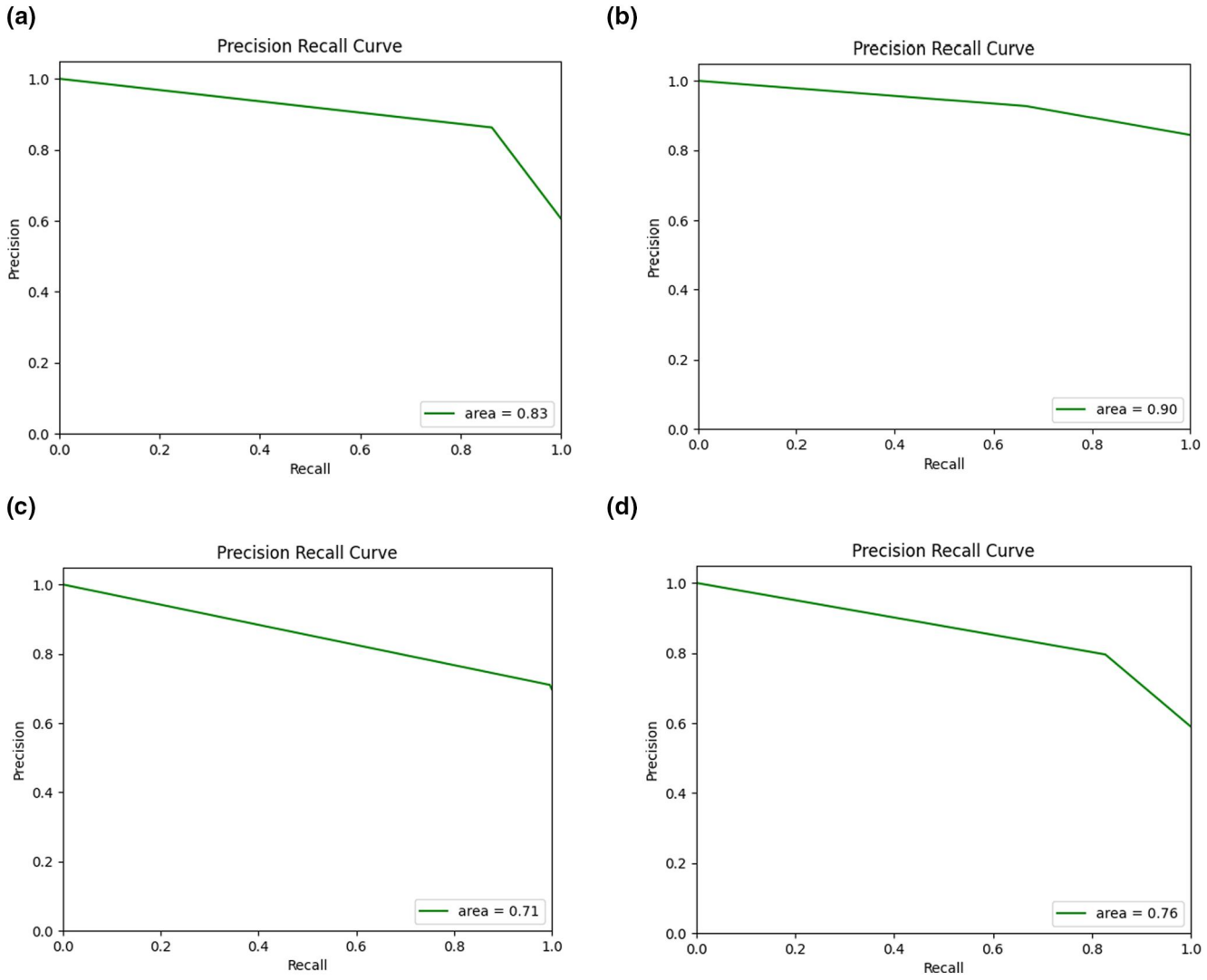

FIGURE 7 (a) D-RF-RS PR curve (INTC). (b) D-RF-RS PR curve (AMD). (c) D-RF-RS PR curve (ATVI). (d) D-RF-RS PR curve (GOOG)

Compared with some previous studies, this study systematically deals with the whole experimental process, exponentially smoothes the time series, and quantifies the practical significance of technical indicators for model training by sorting the importance of features. Random search can find the optimal combination of parameters in a certain range. The joint use of these methods has become the main contribution of this research.

When the accuracy rate exceeds the random probability, it shows that the model can predict the rise and fall of individual stocks to a certain extent, while the stock market is constantly changing and developing, and the stock trend prediction also presents a dynamic development process. In future research, there is still a lot of room for improvement in the selection, screening and use of technical indicators, the selection and optimization of models. This study only uses the stock data of US listed companies. In the future research, experiments can also focus on the trend of US stock index, the reaction of stock index to the market, and the degree of interaction between stock index and individual stocks. In addition, the stock market is constantly

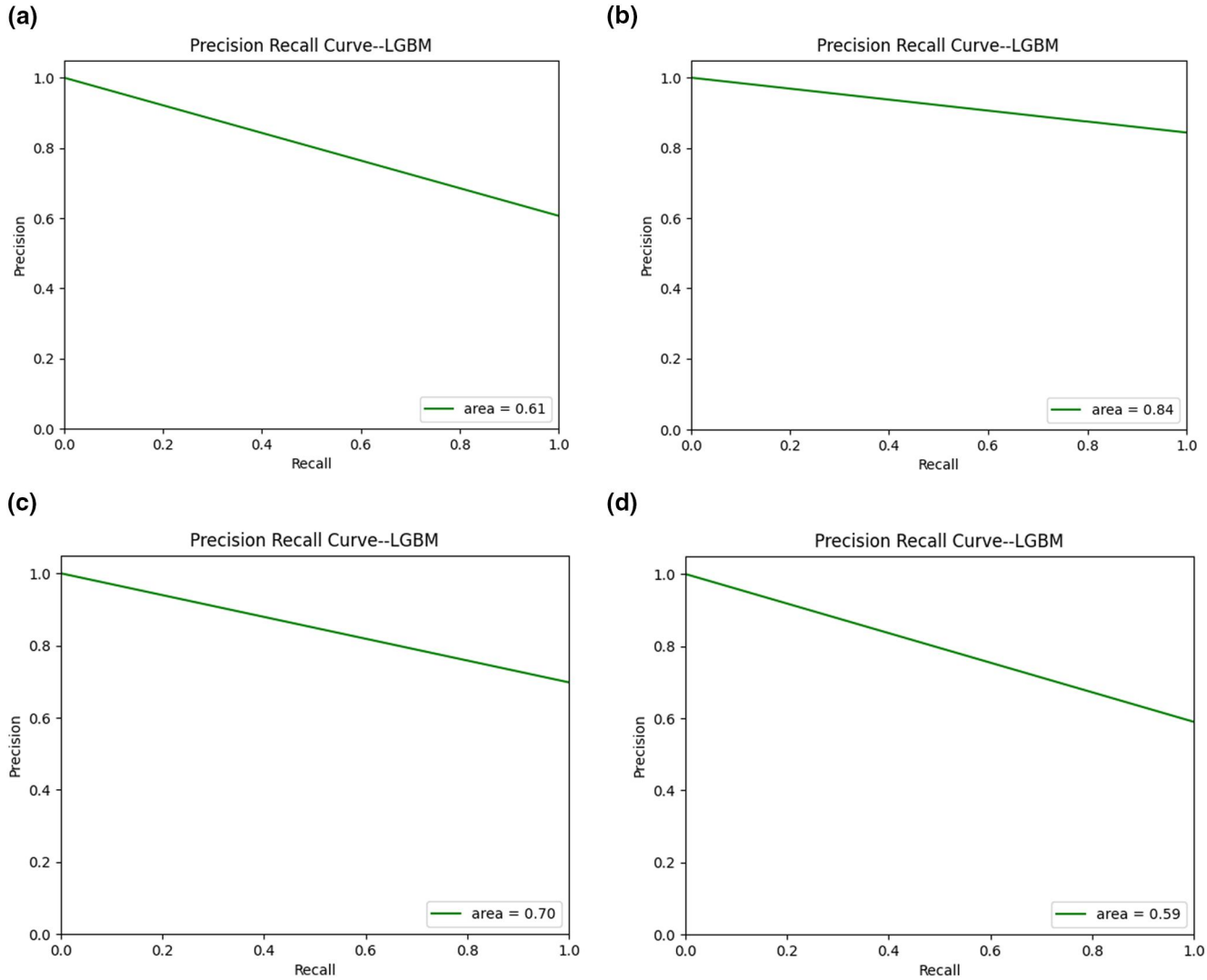

FIGURE 8 (a) LGBM PR curve (INTC). (b) LGBM PR curve (AMD). (c) LGBM PR curve (ATVI). (d) LGBM PR curve (GOOG). LGBM, light gradient boosting machine

developing, and the process of empirical selection and data selection of technical indicators, model selection and optimization still need to be improved. In future research, it is a worthy way to design more novel ensemble learning models, such as the innovation of basic classifiers and different ways of integrating basic classifier.

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CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

The data used to support the findings of this study are available from the corresponding author upon request.

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